# TAO ZHONG

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## EDUCATION

**Princeton University** *08/2023 - 07/2028 (Expected)*
***Degree***: Ph.D. in MAE (Robotics Track) ***cGPA***: 4.0/4.0
***Committee***: Christine Allen-Blanchette (*Chair*), Felix Heide, Ryan P. Adams

**University of Toronto** *09/2018 - 06/2023*
***Degree***: B.A.Sc. in Engineering Science (with High Honour) ***cGPA***: 3.81/4.0
***Major***: Robotics Engineering ***Minor***: Artificial Intelligence
***Advisor***: Animesh Garg

## EXPERIENCE

**CAB Lab**, Princeton University *2023 - Present*
*Graduate Research Student, Advisor: Prof. Christine Allen-Blanchette*
Topics: physics- and geometry-aware deep learning for scientific discovery [C6, C8, U1, U6. U7], robot learning [C4, C5, C7, U3], and generative foundation models [U2, U4, U5].

**People, AI, & Robots Lab**, Vector Institute & University of Toronto *2022 - 2023*
*Undergraduate Research Student, Advisor: Prof. Animesh Garg*
Topics: differentiable physics for dexterous grasp synthesis [C2]

**Noah's Ark Lab**, Huawei Research Canada *2021 - 2022*
*Machine Learning Research Intern, Advisor: Prof. Yang Wang*
Topics: OOD generalization via prompt learning [C3] and test-time adaptation [C1] for foundation models

**aUToronto**, The University of Toronto Self-Driving Car Team *2020 - 2022*
*Mapping & Localization Team Lead, Team Advisors: Prof. Tim Barfoot, Prof. Steven Waslander, Prof. Angela Schoellig, Prof. Jonathan Kelly*
Topics: semantic mapping and SLAM optimization for autonomous driving.

**Shenzhen Institute of Artificial Intelligence and Robotics for Society**, CUHK(SZ) *2020*
*Visiting Research Student, Advisor: Prof. Huihuan Qian*
Topics: web-based testing platform, state estimation, and control for marine robotics

## PUBLICATIONS

* denotes equal contribution.

### Papers in Submission

[U8] **Tao Zhong**, Kaison Fong, Keqin Wang, Christine Allen-Blanchette. Geometric Algebra for Neural Networks: A Survey. *Preprint, 2026.*

[U7] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Symmetry Descent Neural Operators: From Local Equivariance to Global Consistency. *Under Review, 2026.*

[U6] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Homomorphic Generator Learning: Compositional PDE Neural Operators for Multi-Physics Surrogates. *Under Review, 2026.*

[U5] Yixun Hu, Zhixuan Zhao, **Tao Zhong**, Christine Allen-Blanchette. OrbitRIR: Frame-Averaged Few-Shot RIR Synthesis for 3D Sound Source Localization. *Under Review, 2026.*

[U4] Yixun Hu, Zhixuan Zhao, **Tao Zhong**, Christine Allen-Blanchette. CERPA: Heading-Equivariant Attention for Yaw-Invariant Few-Shot Room-Acoustic Synthesis. *Under Review, 2026.*

[U3] Keqin Wang, **Tao Zhong**, Yulong Yang, Christine Allen-Blanchette. Equivariance Begins at the Source: Characterizing Compatible Sources for Generative Robot Policies. *Under Review, 2026.*

[U2] **Tao Zhong**, Dongzhe Zheng, Christine Allen-Blanchette. HodgeCover: Higher-Order Topological Coverage Drives Compression of Sparse Mixture-of-Experts. *Preprint, 2026*.

[U1] Zhixuan Zhao*, **Tao Zhong***, Yixun Hu, Nathalie P. de Leon, Christine Allen-Blanchette. Neural Fields for NV-Center Inverse Sensing. *Preprint, 2026*.

**Refereed Conference Proceedings**

[C8] **Tao Zhong**, Yixun Hu, Dongzhe Zheng, Aditya Sood, Christine Allen-Blanchette. Neural Field Thermal Tomography: A Differentiable Physics Framework for Non-Destructive Evaluation. In *Advances in Neural Information Processing Systems (NeurIPS), 2026*.

[C7] Keqin Wang*, **Tao Zhong***, David Chang, Christine Allen-Blanchette. Local-Canonicalization Equivariant Graph Neural Networks for Sample-Efficient and Generalizable Swarm Robot Control. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026*.

[C6] Dongzhe Zheng, **Tao Zhong**, Christine Allen-Blanchette. Topology-Preserving Neural Operator Learning via Hodge Decomposition. In *Proceedings of the International Conference on Machine Learning (ICML), 2026*.

[C5] **Tao Zhong**, Jonah Buchanan, Christine Allen-Blanchette. Grasp2Grasp: Vision-Based Dexterous Grasp Translation via Schrödinger Bridges. In *Advances in Neural Information Processing Systems (NeurIPS), 2025*.

[C4] **Tao Zhong** and Christine Allen-Blanchette. GAGrasp: Geometric Algebra Diffusion for Dexterous Grasping. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2025*.

[C3] Zhixiang Chi*, Li Gu*, **Tao Zhong**, Huan Liu, Yuanhao Yu, Konstantinos N Plataniotis, Yang Wang. Adapting to Distribution Shift by Visual Domain Prompt Generation. In *Proceedings of the International Conference on Learning Representations (ICLR), 2024*.

[C2] Dylan Turpin, **Tao Zhong**, Shutong Zhang, Guanglei Zhu, Eric Heiden, Miles Macklin, Stavros Tsogkas, Sven Dickinson, Animesh Garg. Fast-Grasp'D: Dexterous Multi-finger Grasp Generation Through Differentiable Simulation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2023*.

[C1] **Tao Zhong***, Zhixiang Chi*, Li Gu*, Yang Wang, Yuanhao Yu, Jin Tang. Meta-DMoE: Adapting to Domain Shift by Meta-Distillation from Mixture-of-Experts. In *Advances in Neural Information Processing Systems (NeurIPS), 2022*.

**Refereed Non-archival Publications**

[W2] Keqin Wang, **Tao Zhong**, Christine Allen-Blanchette. Equivariance Begins at the Source: Quantifying and Characterizing Source Distributions for Generative Robot Policies. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Geometric Representations in Robotics, 2026*.

[W1] **Tao Zhong** and Christine Allen-Blanchette. Geometric Algebra Grasp Diffusion for Dexterous Manipulators. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Equivariant Robotics, 2024*.

**Patents**

[P1] Zhixiang Chi, Li Gu, **Tao Zhong**, Yuanhao Yu, Yang Wang, Jin Tang. Systems and Methods for Artificial-intelligence Model Training Using Unsupervised Domain Adaptation with Multi-source Meta-distillation. *US Patent Application No. 17/966,568*.

## RESEARCH SOFTWARE

**NEFI: Neural-Field Inversion with Differentiable Physics** [Docs] [Code]
A PyTorch toolkit for inverse imaging and sensing with neural fields and differentiable physics, unifying reconstruction, benchmarking, and diagnostics across modalities without training data.

## AWARDS & HONORS

**Lambda Research Grant** *2026*
**Princeton MAE Second Year Departmental Fellowship** (2 / 25+) *2024*
**Princeton University First Year Fellowship in Natural Sciences and Engineering** *2023*
**NeurIPS 2022 Scholar Award** *2022*

**SAE Autodrive Challenge: 1st Place Winner** (As a team) *2020, 2021, 2022*
**University of Toronto Dean's Honours List** (All 8 terms) *2018 - 2023*

## INVITED TALKS

**Vision-Based Dexterous Grasp Translation via Schrödinger Bridges**
GRASP Lab, UPenn, Oct 2025

## TEACHING

**MAE 206 Introduction to Engineering Dynamics** (Undergraduate) *Spring 2026*
Teaching Assistant, Princeton University
**MAE 433 Automatic Control Systems** (Undergraduate) *Fall 2025, Fall 2026*
Teaching Assistant, Princeton University

## MENTORING

**Jonah Buchanan** (with Christine Allen-Blanchette) *2024 - 2025*
**David Chang** (with Christine Allen-Blanchette and Keqin Wang) *2024 - 2025*
**Kaison Fong** (with Christine Allen-Blanchette) *2025*
**Zhixuan Zhao** (with Christine Allen-Blanchette) *2026*

## SERVICE

**Conference Refereeing**
AAAI Conference on Artificial Intelligence (AAAI) *2027*
International Conference on Machine Learning (ICML) *2026*
Neural Information Processing Systems (NeurIPS) *2025 - 2026*
International Conference on Learning Representations (ICLR) *2025 - 2027*
Annual Learning for Dynamics & Control Conference (L4DC) *2024, 2026*

**Journal Refereeing**
IEEE Robotics and Automation Letters (RA-L) *2025*
Transactions on Machine Learning Research (TMLR) *2025 - 2026*

## SKILLS

**Programming Languages:** Python, C/C++, MATLAB/Simulink, Bash, SQL, Assembly
**Libraries & Tools:** PyTorch, NumPy, OpenCV, scikit-learn, ROS, Git, Docker, LaTeX